

plinthreport

Site Intelligence Platform

DEMO — ILLUSTRATIVE DATA ONLY

SITE INTELLIGENCE REPORT

11822 E 116th St N

Collinsville, OK 74021

36.04700°N, 95.81289°W | Rogers County, Oklahoma

[Site Location Map — 1-Mile Context
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REPORT DATE

March 9, 2026

REPORT ID

PLN-20260309-DEMO

PREPARED FOR

— Demo Report —

IECC CLIMATE ZONE

Zone 3 — Mixed-Humid

IMPORTANT — PLEASE READ

This report aggregates publicly available data for informational purposes only. It is not an engineering study, environmental assessment, appraisal, flood insurance determination, or legal opinion. Data values are sourced from federal and state agencies and are presented without interpretation, rating, or recommendation. Site-specific conditions may differ materially from the values shown. This report does not evaluate the suitability of the subject site for any particular use. Always engage qualified licensed professionals — including engineers, surveyors, environmental consultants, and legal counsel — before making development, investment, or financing decisions. The operator of this service assumes no liability for errors, omissions, or reliance on report contents. See Section 8 and the Methodology Appendix for data sources, vintages, and known limitations.

Executive Summary

Site Identification

ADDRESS	11822 E 116th St N Collinsville, OK 74021
COUNTY	Rogers County, Oklahoma
COORDINATES	36.04700°N, 95.81289°W
IECC ZONE	Zone 3 – Mixed-Humid
POP. (5 MI)	42,500 (est.)

Key Observations

FEMA Flood Zone X – area of minimal flood hazard per NFHL mapping.
Seismic PGA 0.05g – low seismic hazard. See Section 2 for details.
Wildfire Hazard Potential: Low (1,240 / 144,153 national index).
NOAA station 11.2 mi away – values may not reflect site microclimate.

Risk & Context Summary

Tabular data summary. Values presented as reported by source datasets without modification. No risk rating, scoring, or suitability determination is made by this report.

Category	Reported Value	Source
Flood Risk	Zone X – Minimal	FEMA NFHL
Seismic PGA	0.05g (MCE _r)	USGS NEHRP 2020
Wildfire WHP	1,240 / 144,153	USDA Forest Service 2023
Hydrologic Group	B/C – Moderate runoff	USDA SSURGO
Air Quality	PM _{2.5} 8.2 µg/m³ (NAAQS: 9.0)	EPA AQ5 2023
FEMA NRI Score	14.2 – Relatively Low	FEMA National Risk Index

SECTION 2

Environmental Risk

Flood Risk – FEMA National Flood Hazard Layer

FLOOD ZONE	Zone X – Area of Minimal Flood Hazard
SFHA (100-YR)	No – not within Special Flood Hazard Area per NFHL
BASE FLOOD ELEV.	Not assigned for Zone X
NFHL DATE	2019-11-15
MAPPED AREA	Yes – FEMA flood mapping available for this area

FEMA flood zone status reflects the NFHL map effective 2019-11-15. Map amendments (LOMA/LOMR), remapping projects, or local ordinance changes may affect current status. Verify current FIRM panel and insurance requirements with your lender, local floodplain administrator, and a licensed flood specialist before relying on this data. Approximately 15-20% of the US has not been mapped by FEMA – unmapped areas are explicitly noted in this report and must never be assumed to be Zone X. See FIRMette (<https://msc.fema.gov>) for the authoritative current map.

Seismic Risk – USGS NEHRP 2020 Seismic Hazard Maps

PGA (MCE _r , 2%/50YR)	0.047g – Very Low
S ₀ (0.2S SPECTRAL)	0.112g
S ₁ (1.0S SPECTRAL)	0.044g
HISTORICAL EQ ACTIVITY	4 events ≥M3.0 within 50 mi, last 50 years. Max magnitude: M4.2.

About Peak Ground Acceleration (PGA)

PGA is the maximum expected ground acceleration at a site, expressed as a fraction of gravitational acceleration (g). It reflects a 2% probability of exceedance in 50 years (the MCE_r level used in building codes). National range and general hazard tiers:

PGA Range	General Hazard Level	Typical Regions
< 0.05g	Very Low	Central & eastern plains, much of the South
0.05 – 0.15g	Low	Midwest, mid-Atlantic, parts of the Southeast
0.15 – 0.30g	Moderate	Central US, Intermountain West
0.30 – 0.60g	High	Pacific Northwest, parts of Utah/Nevada
> 0.60g	Very High	Coastal California, Cascadia subduction zone

These tiers are general context only and do not constitute a seismic design classification. Seismic Design Category (SDC) per ASCE 7 requires a licensed structural engineer. For interactive maps and full methodology, see <https://earthquake.usgs.gov/hazards/>.

Seismic Design Category (SDC) is an engineering determination per ASCE 7 and requires a licensed structural engineer. Hazard values are probabilistic estimates from the USGS national hazard model and do not constitute a site-specific geotechnical evaluation. Source: [USGS NEHRP 2020 Seismic Hazard Maps](#).

Wildfire Hazard Potential – USDA Forest Service

WHP INDEX VALUE **1240 – Low** (national max: 144153)

DATA VINTAGE 2023 (270m resolution)

About Wildfire Hazard Potential (WHP)

WHP is a continuous index produced by the USDA Forest Service Fire Modeling Institute. It reflects the relative potential for wildfire that would be difficult to control, based on fire weather, fuel conditions, and topography. It is *not* a probability of fire occurrence. The continuous index is typically grouped into five general classes:

WHP Range (approx.)	Class
1 – 2,249	Very Low
2,250 – 8,999	Low
9,000 – 35,999	Moderate
36,000 – 143,999	High
≥ 144,000	Very High

Class breakpoints are approximate and may vary by dataset edition. For full methodology and downloads, see <https://www.firelab.org/project/wildfire-hazard-potential>.

WHP is a modeled index, not a regulatory classification. Wildland-Urban Interface (WUI) status and local fire hazard severity zones (FHSZ) may impose additional construction and insurance requirements. Consult your local fire authority and AHJ. Source: [USDA Forest Service Fire Modeling Institute, 2023](#).

Soil Profile – USDA SSURGO

MAP UNIT Dennis-Pharoah complex, 1 to 3 percent slopes

HYDROLOGIC GROUP B/C

DRAINAGE CLASS Well drained

DOMINANT COMPONENT Dennis (55.0%)

TAXONOMIC CLASS *Fine, mixed, active, thermic Oxyaquic Hapludalfs*

NOTE Nearest mapped unit returned – site may be in an unmapped area (e.g., active floodplain). Verify with USDA Web Soil Survey.

Soil data is presented as raw USDA SSURGO taxonomy. No foundation suitability, bearing capacity, or construction assessment is made or implied by this report. Consult a licensed geotechnical engineer for foundation design and site-specific soil investigation. Source: USDA NRCS SSURGO database.

Air Quality – EPA Air Quality System (AQS)

EPA AQS credentials not yet configured. Data unavailable for this demo.

EPA AQS data typically lags 6-18 months from current conditions and is measured at fixed monitoring stations that may not reflect site-level air quality. This is not a real-time air quality index. No health assessment is made or implied. Source: EPA Air Quality System.

FEMA National Risk Index – Composite Hazard Risk Score

Attribution required by source: The values below are produced by FEMA's National Risk Index model – a composite score developed and published by FEMA. This is FEMA's model output, not a rating derived by Plinth. See <https://hazards.fema.gov/nri/> for full methodology. Source: FEMA National Risk Index.

CENSUS TRACT 40131950200

NRI RISK SCORE **14.2 – Relatively Low**

About the FEMA National Risk Index Score

The NRI composite score is a percentile-based value (0-100) representing a census tract's relative natural hazard risk compared to all other US tracts, incorporating expected annual loss, social vulnerability, and community resilience. FEMA uses five rating categories:

Score Range	FEMA Rating
0 – 20	Very Low
20 – 40	Relatively Low
40 – 60	Relatively Moderate
60 – 80	Relatively High
80 – 100	Very High

Component hazard scores use the same 0-100 percentile scale relative to other US tracts. A higher score indicates higher relative risk for that hazard compared to the national distribution. For full methodology, data dictionary, and interactive map, see <https://hazards.fema.gov/nri/>.

Hazard Component Scores (FEMA NRI – Source: FEMA, not Plinth-derived)

Hazard	NRI Score (0-100 percentile)
Riverine Flooding	9.8
Tornado	21.4
Hail	18.7
Strong Wind	11.2
Winter Weather	7.3
Wildfire	3.1
Earthquake	2.4

Physical Context

Elevation – USGS 3DEP (10m)

SITE ELEVATION	693.4 ft (211.3 m)
DEM RESOLUTION	10.0m (1/3 arc-second)

Slope – USGS 3DEP Derived

MEAN SLOPE	1.9%
MAX SLOPE	3.6%
RADIUS	100m
REFERENCE BANDS	0-5% / 5-15% / >15%

Slope values are DEM-derived estimates (100m radius). Site-specific grading conditions should be confirmed by a licensed surveyor or civil engineer.

[Terrain / Hillshade Map
1-Mile Radius
Scale bar · North arrow
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Land Cover – USGS NLCD 2021 (500m radius)

DOMINANT CLASS **Developed, Low Intensity**

NLCD CODE 22

IMPERVIOUS SURFACE 31.2% (estimated)

Land Cover Class	% of Radius
Developed, Low Intensity	41.8%
Grassland/Herbaceous	27.6%
Pasture/Hay	19.3%
Cultivated Crops	6.7%
Open Water	4.6%

Hydrography – NHDPlus High Resolution

NEAREST WATERBODY Verdigris River (LakePond) – 4.24 mi

NEAREST FLOWLINE Bird Creek (StreamRiver) – 1.33 mi

SEARCH RADIUS 5 miles

Climate & Weather

CLIMATE STATION	TULSA INTERNATIONAL AIRPORT (USC00348818)
DISTANCE FROM SITE	11.2 miles
NORMALS PERIOD	NOAA 1991-2020 Climate Normals
IECC ZONE	Zone 3 – Mixed-Humid. Local jurisdiction adoption may vary. Verify with AHJ.

3142 HDD ANNUAL	2218 CDD ANNUAL	64% AVG REL. HUMIDITY	62 FREEZE-THAW DAYS
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Monthly Climate Normals (1991-2020)

Metric	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Max Temp (°F)	48	54	63	73	80	89	94	94	85	73	60	49
Min Temp (°F)	28	32	40	49	59	68	72	71	62	50	39	29
Precip (in)	1.73	2.06	3.14	3.81	5.42	4.28	3.10	2.87	4.21	4.08	2.84	2.15
Snowfall (in)	1.4	1.8	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	1.1

Metric Definitions

Heating Degree Days (HDD)

The sum of degrees per day that the mean temperature falls below 65°F over a year. A day averaging 45°F contributes 20 HDD. Used to estimate heating energy demand – higher HDD = more heating needed.

Cooling Degree Days (CDD)

The sum of degrees per day that the mean temperature exceeds 65°F over a year. A day averaging 85°F contributes 20 CDD. Used to estimate cooling/AC energy demand.

Freeze-Thaw Days

The number of days per year where the temperature crosses the 32°F (0°C) threshold – either rising above or dropping below freezing within the same day. Relevant for pavement and concrete durability, since repeated freeze-thaw cycles cause cracking and spalling.

Climate values are NOAA 1991-2020 climate normals measured at TULSA INTERNATIONAL AIRPORT, 11.2 miles from the site. Station-based normals may not reflect site-level microclimatic variation due to local topography, urban heat island effects, or distance from station. Source: NOAA National Centers for Environmental Information.

Solar Resource

ANNUAL AVG GHI4.83 kWh/m²/day peak sun hours

LATITUDE36.047°N

Monthly Solar Resource (GHI, kWh/m²/day)

Metric	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
GHI	2.71	3.42	4.38	5.41	6.12	6.88	6.73	6.21	5.11	4.02	2.83	2.41
Sunrise	7:32	7:06	6:25	6:37	6:01	5:50	5:58	6:26	6:57	7:27	7:00	7:28
Sunset	5:32	6:03	6:33	7:01	7:29	8:00	8:10	7:46	7:01	6:17	5:40	5:26



Solar resource values are long-term climatological averages (NASA POWER, 2001–2020) at the grid cell containing this location (≈0.5° resolution). They represent horizontal surface irradiance and do not account for shading, panel tilt, azimuth, or system losses. No energy production estimate is made or implied. Source: [NASA POWER \(https://power.larc.nasa.gov/\)](https://power.larc.nasa.gov/), accessed March 9, 2026.

Demographic & Market Context

Area-weighted block group intersections. Straight-line radius buffers – physical barriers not accounted for. Source: 2019–2023 ACS 5-Year Estimates.

Variable	1-Mile Radius	5-Mile Radius	10-Mile Radius
POPULATION & AGE			
Total Population	8,200	42,500	128,000
Median Age	36.2	35.8	36.4
Average Household Size	2.61	2.63	2.57
HOUSEHOLDS & INCOME			
Total Households	3,150	16,800	51,200
Median Household Income	\$62,400	\$68,200	\$61,800
Per Capita Income	\$31,100	\$33,400	\$29,200
% Below Poverty Level	10.1%	9.2%	12.4%
% Receiving Public Assistance	2.8%	2.4%	3.6%
HOUSING STOCK			
Total Housing Units	3,410	18,100	56,400
Median Home Value	\$181,000	\$196,000	\$174,000
Median Gross Rent	\$854	\$924	\$882
Vacancy Rate	7.6%	7.2%	9.1%
Median Year Built	1991	1994	1988
TENURE			
% Owner-Occupied	71%	69%	65%
% Renter-Occupied	29%	31%	35%
EDUCATION			
% Bachelor's Degree or Higher	28%	31%	29%
% High School Diploma or Higher	89%	90%	87%
COMMUTE & EMPLOYMENT			
Mean Travel Time to Work (min)	28.4	26.1	28.7
% Drive Alone	84.8%	84.2%	83.1%
% Work from Home	7.9%	10.2%	9.1%

Variable	1-Mile Radius	5-Mile Radius	10-Mile Radius
Unemployment Rate	4.2%	3.8%	4.5%
Labor Force Participation Rate	61.8%	63.1%	61.2%
RACE & ETHNICITY			
% Non-Hispanic White	78.2%	76.4%	72.1%
% Hispanic or Latino	7.8%	9.1%	11.3%
% Black or African American	5.2%	6.0%	8.1%
% Asian	2.1%	2.4%	2.8%
% Other / Multiracial	6.7%	6.1%	5.7%
LANGUAGE			
% Limited English Proficiency	2.9%	3.8%	5.1%

Demographic values are area-weighted estimates derived from Census block groups intersecting straight-line radius buffers. Physical barriers (rivers, highways, etc.) are not accounted for. Population counts may differ from administrative area totals. Radius buffers use geodesic measurement. All values are estimates with associated margins of error per ACS methodology – consult source tables for confidence intervals. These data are presented for general context only and must not be used as a substitute for a formal market study or demographic analysis. Source: 2019–2023 ACS 5-Year Estimates.

Infrastructure & Access

Broadband Availability – FCC National Broadband Map

Required disclosure: FCC broadband data is self-reported by internet service providers and may significantly overstate actual service coverage and speeds at any given location. Treat these values as indicative only. Verify broadband availability and tier directly with each listed service provider before relying on this data for site selection, feasibility analysis, or development planning. Source: FCC National Broadband Map, deployment data.

Provider	Technology	Max Down (Mbps)	Max Up (Mbps)
Metronet	Fiber	5000	5000
Cox Communications	Cable	1000	35
AT&T	Fiber	1000	1000
T-Mobile	LTE	100	20
Viasat	Satellite	150	12

Road & Transportation Access

Site is approximately 0.3 miles from US-20 (East Admiral Place) and 4.1 miles from US-412.

No fixed-route public transit service identified within 5 miles of this location.

Water & Sewer

Verify water and sewer service availability directly with the City of Collinsville Public Works and Rogers County Rural Water District. No authoritative national data source for municipal water/sewer service area boundaries is currently available in this platform.

Data Sources & Methodology

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Data Source	Version / Date	Refresh Cadence	Storage
FEMA National Flood Hazard Layer (NFHL)	2019-11-15	Monthly	PostGIS
USGS 3D Elevation Program (3DEP)	2023	Annual	MinIO COG
USGS National Land Cover Database (NLCD)	2021	Annual	MinIO COG
USGS NEHRP 2020 Seismic Hazard	2020	API (365d TTL)	query_cache
USGS Earthquake Catalog (FDSN)	Real-time	API (30d TTL)	query_cache
USDA NRCS SSURGO	2024-03	Annual	PostGIS
USDA Forest Service Wildfire Hazard Potential	2023	Annual	API (365d TTL)
FEMA National Risk Index	2023	Annual	PostGIS
US Census TIGER/Line Block Groups	2023	Annual	PostGIS
US Census ACS 5-Year Estimates	2019-2023	Annual	API (365d TTL)
IECC Climate Zone Map (DOE/PNNL)	IECC 2021	Per code cycle	PostGIS
NOAA US Climate Normals 1991-2020	2020	Decadal	PostGIS
NASA POWER Climatology 2001-2020	2020	API (30d TTL)	query_cache
NHDPlus High Resolution Hydrography	2023	Annual	PostGIS
FCC National Broadband Map	2024-06	Semi-annual	PostGIS
EPA Air Quality System (AQS)	2023	Annual	query_cache

About This Report

What Plinth Is

Plinth is a site intelligence platform that aggregates publicly available federal and state datasets into a single, structured PDF report for a given US address. Reports are generated on demand and are intended to provide an organized starting point for predevelopment due diligence.

What This Report Is Not

This report is not an engineering study, Phase I or Phase II environmental site assessment, appraisal, wetlands delineation, title report, flood insurance determination, air quality permit review, geotechnical investigation, or legal opinion. Nothing in this report constitutes professional engineering, surveying, environmental, or legal advice. All interpretations, determinations, and recommendations regarding site suitability, development feasibility, risk management, or regulatory compliance require engagement of appropriately licensed professionals.

Data Accuracy & Limitations

All data values are sourced directly from the cited federal or state datasets and are presented without modification or interpretation. Known limitations of each dataset are described inline in the relevant report section. Dataset vintages vary; see Section 8 for effective dates and retrieval timestamps. Plinth does not verify real-time accuracy of source data.

No Risk Rating System

This report does not employ any color-coded risk rating, composite suitability score, or editorial judgment about site quality. All values are raw source data. FEMA National Risk Index scores are reproduced verbatim from FEMA's published model and are explicitly attributed to FEMA (see Section 2).

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A snapshot of data source versions in effect at generation time is archived with this report's unique ID (PLN-20260309-DEMO). Archived reports represent conditions as of the report generation date and are not updated after delivery.

This is a demonstration report. All values are illustrative. Do not rely on this report for any purpose.